9.1 Factoring Quadratic Polynomials - Worksheet 1

1 Translate the diagram of algebra tiles into an equation.

x^2	x	x	x	x	x
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1

2 Use a diagram of algebra tiles to factor $x^2 + 8x + 15$. Draw the diagram and write the final equation.

You may want to do some actual scratch work on scratch paper for this.

3 Fill in the appropriate value into the boxes, then use the ac method to factor the given quadratic polynomial using a complete presentation.

$$x^2 + 8x + 15 \longrightarrow \begin{cases} \text{Multiply to } \boxed{} \\ \text{Add to } \boxed{} \end{cases}$$

9.2 Factoring Quadratic Polynomials - Worksheet 2

1 Use a diagram of algebra tiles to factor $2x^2 + 9x + 4$. Draw the diagram and write the final equation.

Put both of your x^2 tiles on the top left corner.

2 Fill in the appropriate value into the boxes, then use the ac method to factor the given quadratic polynomial using a complete presentation.

$$x^2 + 2x - 8 \longrightarrow \begin{cases} \text{Multiply to } \boxed{} \\ \text{Add to } \boxed{} \end{cases}$$

3 Fill in the appropriate value into the boxes, then use the ac method to factor the given quadratic polynomial using a complete presentation.

$$2x^2 - 3x - 28 \longrightarrow \begin{cases} \text{Multiply to } \boxed{} \\ \text{Add to } \boxed{} \end{cases}$$

9.3 Factoring Quadratic Polynomials - Worksheet 3

1 Fill in the appropriate value into the boxes, then use the ac method to factor the given quadratic polynomial using a complete presentation.

$$x^2 - 8x + 7 \longrightarrow \begin{cases} \text{Multiply to } \boxed{} \\ \text{Add to } \boxed{} \end{cases}$$

2 Fill in the appropriate value into the boxes, then use the ac method to factor the given quadratic polynomial using a complete presentation.

$$2x^2 + 5x + 2 \longrightarrow \begin{cases} \text{Multiply to } \boxed{} \\ \text{Add to } \boxed{} \end{cases}$$

3 Use the ac method to factor $x^2 - 7x + 10$ using a complete presentation.

You can continue to write out the two conditions if you want. Eventually, that will be a mental exercise

9.4 Factoring Quadratic Polynomials - Worksheet 4

1 Fill in the appropriate value into the boxes, then use the ac method to factor the given quadratic polynomial using a complete presentation.

$$3x^2 - 10x - 8 \longrightarrow \begin{cases} \text{Multiply to } \boxed{} \\ \text{Add to } \boxed{} \end{cases}$$

2 Use the ac method to factor $x^2 + 6x + 9$ using a complete presentation.

3 Use the ac method to factor $x^2 - 3x - 40$ using a complete presentation.

4 Use the ac method to factor $2x^2 - 5x - 3$ using a complete presentation.

9.5 Factoring Quadratic Polynomials - Worksheet 5

1 Use the ac method to factor $x^2 + 9x + 20$ using a complete presentation.

2 Use the ac method to factor $4x^2 - 4x - 3$ using a complete presentation.

3 Factor $x^2 + 10x + 16$ and $x^2 - 10x + 16$, then compare the results. What do you notice about the factorizations?

Looking for patterns is a core element of mathematical thinking.

4 Factor $x^2 - 5x - 14$ and $x^2 + 5x - 14$, then compare the results. What do you notice about the factorizations?